LAB 6

NAME: Manuel Castillo Obregón

ID: u269423

GitHub Labs repository: https://github.com/manueleco/ml-for-sound-and-music.git

Google Colab URL for the lab: https://drive.google.com/file/d/12PSV3WsZU9DG0uIrVPHZQSRoMYx-6DNm/view?usp=sharing

SETUP

- The notebook was executed in a remote server (Modal) so a try catch was implemented so we could decide about resources and paths depending on the environment: MacBook with M1 chip, Google Colab, Modal or any PC with Cuda (nvidia) cores available to use the GPU to its maximum.
- Installed the required dependencies for PyTorch training, audio I/O, visualization, perceptual/spectral losses, and CO2 tracking.
- Detected the available backend (CUDA or MPS or CPU) and set random seedsto make it global.
- Defined the main experiment params: 16 kHz sample rate and fixed 1 second clips (16,000 samples) to keep training consistent and efficient.

In [1]:

```python
import os, sys, zipfile, random, time, pathlib
import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
import torchaudio

try:
    import soundfile as sf
except Exception:
    !pip -q install soundfile
    import soundfile as sf

from torch.utils.data import Dataset, DataLoader
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import trange
from pathlib import Path

try:
    from codecarbon import EmissionsTracker
    USE_CODECARBON = True
except Exception:
    !pip -q install codecarbon
    from codecarbon import EmissionsTracker
    USE_CODECARBON = True

CODECARBON_DIR = Path("./codecarbon_logs")
CODECARBON_DIR.mkdir(parents=True, exist_ok=True)

try:
    import psutil
    USE_PSUTIL = True
except Exception:
    !pip -q install psutil
    import psutil
    USE_PSUTIL = True

import subprocess

print("PWD:", pathlib.Path().resolve())
print("Root:", list(pathlib.Path("/mnt").glob("*")))

if torch.cuda.is_available():
    device = "cuda"
elif torch.backends.mps.is_available():
    device = "mps"
else:
    device = "cpu"
print("Device:", device)

torch.set_num_threads(max(1, (os.cpu_count() or 4) - 1))

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
if device == "cuda":
    torch.cuda.manual_seed_all(SEED)

SAMPLE_RATE = 16000
TARGET_LENGTH = 16000
BATCH_SIZE = 64
N_EPOCHS = 50
N_EPOCHS_PERC = 50

DATA_ROOT = None

proc = psutil.Process(os.getpid()) if USE_PSUTIL else None

def _bytes_to_gb(x):
    return float(x) / (1024**3)

def get_cpu_ram():
    if proc is None:
        return None, None
    try:
        cpu = psutil.cpu_percent(interval=None)
        rss = _bytes_to_gb(proc.memory_info().rss)
        return float(cpu), float(rss)
    except Exception:
        return None, None

def get_gpu_mem():
    if device != "cuda":
        return None, None
    try:
        alloc = _bytes_to_gb(torch.cuda.memory_allocated())
        peak = _bytes_to_gb(torch.cuda.max_memory_allocated())
        return float(alloc), float(peak)
    except Exception:
        return None, None
```

```
[notice] A new release of pip is available: 24.3.1 -> 25.3
[notice] To update, run: pip install --upgrade pip

[notice] A new release of pip is available: 24.3.1 -> 25.3
[notice] To update, run: pip install --upgrade pip
PWD: /root
Root: [PosixPath('/mnt/mtat-datalabs')]
Device: cuda
```

DATA LOADING WITH MAGNATAGATUNE

Reutilizamos parcialmente el codigo del lab anterior

En lab 5 ya teniamos:

- Mirdata initialize
- Un dataset
- Segmentación

Implemented a robust loading strategy that works across environments:

- Modal Volume (if the dataset is mounted at /mnt/mtat-datalabs)
- mirdata (if the dataset is available in the installed mirdata version)
- Local fallback (searching for annotations_final.csv and clip_info_final.csv)

Summary: Built train/validation/test splits (80/10/10) and optionally capped the number of examples to control training time and cost (mainly when in need for it to execute fast)

In [2]:

```python
# intento de carga con mirdata y fallback a versión local o modal

USE_MIRDATA = False

# detectar Modal

IS_MODAL = bool(os.environ.get("MODAL_TASK_ID") or os.environ.get("MODAL_ENVIRONMENT") or os.environ.get("MODAL_RUN_ID"))
# ruta real modal
MODAL_VOLUME_ROOT = Path("/mnt/mtat-datalabs")

def read_csv_flexible(path):
    try:
        return pd.read_csv(path, sep="\t")
    except Exception:
        return pd.read_csv(path, sep=",")

def make_splits_from_dfs(df_ann, df_info, audio_root, max_tracks=10000):
    df_ann.columns = [c.strip() for c in df_ann.columns]
    df_info.columns = [c.strip() for c in df_info.columns]

    if "clip_id" not in df_ann.columns:
        raise ValueError(f"annotations_final.csv missing 'clip_id'. Columns: {list(df_ann.columns)[:20]}")

    if "clip_id" not in df_info.columns or "mp3_path" not in df_info.columns:
        raise ValueError(f"clip_info_final.csv must contain 'clip_id' and 'mp3_path'. Columns: {list(df_info.columns)[:20]}")

    all_ids = df_ann["clip_id"].values
    if len(all_ids) > max_tracks:
        all_ids = all_ids[:max_tracks]

    all_ids = list(all_ids)
    random.shuffle(all_ids)

    n_total = len(all_ids)
    n_train = int(0.8 * n_total)
    n_val = int(0.1 * n_total)

    train_ids = all_ids[:n_train]
    val_ids = all_ids[n_train:n_train + n_val]
    test_ids = all_ids[n_train + n_val:]

    df_info_indexed = df_info.set_index("clip_id")

    def make_df(id_list):
        rows = []
        for cid in id_list:
            if cid in df_info_indexed.index:
                rows.append({"clip_id": cid, "mp3_path": df_info_indexed.loc[cid, "mp3_path"]})
        return pd.DataFrame(rows)

    train_df = make_df(train_ids)
    val_df = make_df(val_ids)
    test_df = make_df(test_ids)

    return train_df, val_df, test_df, Path(audio_root)

if IS_MODAL and (MODAL_VOLUME_ROOT / "annotations_final.csv").exists():
    data_dir = str(MODAL_VOLUME_ROOT)
    ann_path = os.path.join(data_dir, "annotations_final.csv")
    info_path = os.path.join(data_dir, "clip_info_final.csv")

    # normalizar nombres de columnas por si vienen raros

    df_ann = read_csv_flexible(ann_path)
    df_info = read_csv_flexible(info_path)

    audio_root = os.path.join(data_dir, "audio")
    train_df, val_df, test_df, DATA_ROOT = make_splits_from_dfs(df_ann, df_info, audio_root)

    print("Using MagnaTagATune from Modal Volume.")
    print("Sizes (train, val, test):", len(train_df), len(val_df), len(test_df))
    print("DATA_ROOT:", DATA_ROOT)

else:
    try:
        import mirdata

        # algunos installs de mirdata no traen magnatagatune entonces si falla lo llevo al except

        data_home = os.environ.get("MTAT_MIRDATA_HOME", "./data/mtat")
        mtat = mirdata.initialize("magnatagatune", data_home=data_home)
        mtat.download()
        mtat.validate()

        all_tracks_dict = mtat.load_tracks()
        all_tracks = list(all_tracks_dict.values())
        random.shuffle(all_tracks)

        n_total = len(all_tracks)
        n_train = int(0.8 * n_total)
        n_val = int(0.1 * n_total)

        train_tracks = all_tracks[:n_train]
        val_tracks = all_tracks[n_train:n_train + n_val]
        test_tracks = all_tracks[n_train + n_val:]

        USE_MIRDATA = True
        print("Using MagnaTagATune via mirdata.")
        print("Sizes (train, val, test):", len(train_tracks), len(val_tracks), len(test_tracks))

    except Exception as e:
        print("mirdata not available or dataset not supported in this mirdata install. Falling back to local dataset.")
        print("Detail:", e)

        IN_COLAB = "google.colab" in sys.modules
        if IN_COLAB:
            from google.colab import drive
            drive.mount("/content/drive")
            search_roots = [Path("/content/drive/MyDrive"), Path("/content")]
        else:
            search_roots = [Path.cwd(), Path.cwd().parent, Path.cwd().parent.parent]

        data_path = None
        for root in search_roots:
            if root.exists():
                for p in root.rglob("annotations_final.csv"):
                    data_path = p.parent
                    break
            if data_path is not None:
                break

        if data_path is None:
            raise FileNotFoundError("Could not find annotations_final.csv in known locations")

        data_dir = str(data_path)
        ann_path = os.path.join(data_dir, "annotations_final.csv")
        info_path = os.path.join(data_dir, "clip_info_final.csv")

        df_ann = read_csv_flexible(ann_path)
        df_info = read_csv_flexible(info_path)

        audio_root = os.path.join(data_dir, "audio")
        os.makedirs(audio_root, exist_ok=True)

        full_zip = os.path.join(data_dir, "mp3_full.zip")
        if os.path.exists(full_zip) and not os.path.exists(os.path.join(audio_root, "000")):
            with zipfile.ZipFile(full_zip, "r") as z:
                z.extractall(audio_root)

        train_df, val_df, test_df, DATA_ROOT = make_splits_from_dfs(df_ann, df_info, audio_root)

        print("Using local MagnaTagATune.")
        print("Sizes (train, val, test):", len(train_df), len(val_df), len(test_df))
        print("DATA_ROOT:", DATA_ROOT)
```

```
mirdata not available or dataset not supported in this mirdata install. Falling back to local dataset.
Detail: No module named 'mirdata'
Using local MagnaTagATune.
Sizes (train, val, test): 8000 1000 1000
DATA_ROOT: /__modal/volumes/vo-IDPx0DuUwAPUOrIsswHOuI/datalabs/audio
```

Dataset para encoder

AUTOENCODER BASELINE (MSE)

Autoencoder convolucional

Implemented a 1D convolutional autoencoder:

- Encoder with 3 Conv 1D layers (stride=2) to progressively downsample the waveform.
- Fixed size fully connected bottleneck of 512 (compression rate unchanged as required).
- Decoder with ConvTranspose1D layers to reconstruct the waveform back to the original length.

Trained a baseline model using MSE loss to obtain a quantitative reference and reconstruction baseline you can trust in time domain.

In [ ]:

```python
# dataset para autoencoder (mirdata)
class MTATAutoencoderDatasetMirdata(Dataset):
    def __init__(self, tracks, split="train"):
        self.tracks = tracks
        self.split = split

    def __len__(self):
        return len(self.tracks)

    def __getitem__(self, idx):
        # carga de datos
        track = self.tracks[idx]
        audio, sr = track.audio

        # conversión a tensor
        audio = torch.tensor(audio, dtype=torch.float32)

        # conversion a mono
        if audio.ndim > 1:
            audio = audio.mean(dim=0)

        # resample
        if sr != SAMPLE_RATE:
            audio = torchaudio.functional.resample(
                audio.unsqueeze(0),
                orig_freq=sr,
                new_freq=SAMPLE_RATE
            ).squeeze(0)

        # tratamiento de audio largo (como seccion 2)
        audio = self._crop_or_pad(audio)

        return {"audio": audio}

    def _crop_or_pad(self, audio):
        length = audio.shape[-1]
        if length > TARGET_LENGTH:
            if self.split == "train":
                start = random.randint(0, length - TARGET_LENGTH)
            else:
                start = (length - TARGET_LENGTH) // 2
            audio = audio[start:start + TARGET_LENGTH]
        elif length < TARGET_LENGTH:
            pad = TARGET_LENGTH - length
            audio = F.pad(audio, (0, pad))
        return audio



# dataset para autoencoder (local: CSV + mp3)
class MTATAutoencoderDatasetLocal(Dataset):
    def __init__(self, df, split="train", data_root=None):
        self.df = df.reset_index(drop=True)
        self.split = split
        self.data_root = Path(data_root) if data_root is not None else None

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        # carga de datos
        row = self.df.iloc[idx]
        mp3_rel = str(row["mp3_path"])

        # normalización de ruta (evita duplicaciones)
        if mp3_rel.startswith("audio/"):
            mp3_rel = mp3_rel[len("audio/"):]

        audio_path = self.data_root / mp3_rel

        # lectura de audio (evita torchcodec/ffmpeg)
        # (T, C)
        audio_np, sr = sf.read(str(audio_path), always_2d=True)

        # conversión a tensor (C, T)
        audio = torch.tensor(audio_np, dtype=torch.float32).T

        # conversion a mono
        if audio.size(0) > 1:
            audio = audio.mean(dim=0, keepdim=True)

        # resample
        if sr != SAMPLE_RATE:
            audio = torchaudio.functional.resample(
                audio,
                orig_freq=sr,
                new_freq=SAMPLE_RATE
            )
            sr = SAMPLE_RATE

        audio = audio.squeeze(0)

        # tratamiento de audio largo (como seccion 2)
        audio = self._crop_or_pad(audio)

        return {"audio": audio}

    def _crop_or_pad(self, audio):
        length = audio.shape[-1]
        if length > TARGET_LENGTH:
            if self.split == "train":
                start = random.randint(0, length - TARGET_LENGTH)
            else:
                start = (length - TARGET_LENGTH) // 2
            audio = audio[start:start + TARGET_LENGTH]
        elif length < TARGET_LENGTH:
            pad = TARGET_LENGTH - length
            audio = F.pad(audio, (0, pad))
        return audio


# creación de datasets y dataloaders
if USE_MIRDATA:
    train_dataset = MTATAutoencoderDatasetMirdata(train_tracks, split="train")
    val_dataset   = MTATAutoencoderDatasetMirdata(val_tracks,   split="val")
    test_dataset  = MTATAutoencoderDatasetMirdata(test_tracks,  split="test")
else:
    if DATA_ROOT is None:
        raise ValueError("DATA_ROOT no está definido en fallback local")
    train_dataset = MTATAutoencoderDatasetLocal(train_df, split="train", data_root=DATA_ROOT)
    val_dataset   = MTATAutoencoderDatasetLocal(val_df,   split="valid", data_root=DATA_ROOT)
    test_dataset  = MTATAutoencoderDatasetLocal(test_df,  split="test",  data_root=DATA_ROOT)
import os


import platform

IS_MACOS = (platform.system().lower() == "darwin")

if IS_MACOS:
    NUM_WORKERS = 0
else:
    NUM_WORKERS = min(8, (os.cpu_count() or 8))

pin_memory = (device == "cuda")

train_loader = DataLoader(
    train_dataset,
    batch_size=BATCH_SIZE,
    shuffle=True,
    drop_last=True,
    num_workers=NUM_WORKERS,
    pin_memory=pin_memory,
    persistent_workers=(NUM_WORKERS > 0),
    prefetch_factor=2 if NUM_WORKERS > 0 else None
)

val_loader = DataLoader(
    val_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    drop_last=False,
    num_workers=NUM_WORKERS,
    pin_memory=pin_memory,
    persistent_workers=(NUM_WORKERS > 0),
    prefetch_factor=2 if NUM_WORKERS > 0 else None
)

test_loader = DataLoader(
    test_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    drop_last=False,
    num_workers=NUM_WORKERS,
    pin_memory=pin_memory,
    persistent_workers=(NUM_WORKERS > 0),
    prefetch_factor=2 if NUM_WORKERS > 0 else None
)

print("DataLoaders listos:")
print("Train batches:", len(train_loader))
print("Val batches:  ", len(val_loader))
print("Test batches: ", len(test_loader))
print("NUM_WORKERS:", NUM_WORKERS)

t0 = time.time()
_ = next(iter(train_loader))
print("Tiempo para 1 batch:", time.time() - t0, "sec")
```

```
DataLoaders listos:
Train batches: 125
Val batches:   16
Test batches:  16
NUM_WORKERS: 8
Tiempo para 1 batch: 3.0653979778289795 sec
```

In [4]:

```python
# perceptual / spectral loss (auraloss con fallback)
mse_loss = nn.MSELoss()

USE_AURALOSS = False
try:
    import auraloss
    mrstft = auraloss.freq.MultiResolutionSTFTLoss(
        fft_sizes=[512, 1024, 2048],
        hop_sizes=[128, 256, 512],
        win_lengths=[512, 1024, 2048]
    ).to(device)
    USE_AURALOSS = True

    def perceptual_loss(x_rec, x):
        return mrstft(x_rec, x)

except Exception:
    def multi_scale_stft_loss(x, y, scales=(512, 1024, 2048), hop_ratio=0.25, eps=1e-7):
        loss = 0.0
        for n_fft in scales:
            hop = int(n_fft * hop_ratio)
            window = torch.hann_window(n_fft, device=x.device)
            X = torch.stft(x.squeeze(1), n_fft=n_fft, hop_length=hop, window=window, return_complex=True)
            Y = torch.stft(y.squeeze(1), n_fft=n_fft, hop_length=hop, window=window, return_complex=True)
            Xmag = torch.log(X.abs() + eps)
            Ymag = torch.log(Y.abs() + eps)
            loss += torch.mean(torch.abs(Xmag - Ymag))
        return loss / len(scales)

    def perceptual_loss(x_rec, x):
        return multi_scale_stft_loss(x_rec, x)

def combined_loss(x_rec, x, alpha=0.01):
    l_mse = mse_loss(x_rec, x)
    l_stft = perceptual_loss(x_rec, x)
    return l_mse + alpha * l_stft, l_mse, l_stft

print("Perceptual loss:", "auraloss MRSTFT" if USE_AURALOSS else "manual multi-scale STFT")
```

```
Perceptual loss: manual multi-scale STFT
```

In [5]:

```python
# autoencoder convolucional
class AudioAutoencoder(nn.Module):
    def __init__(self, input_length=TARGET_LENGTH, bottleneck_size=512):
        super().__init__()
        self.input_length = input_length
        self.bottleneck_size = bottleneck_size

        # encoder
        self.enc_conv1 = nn.Conv1d(1, 16, kernel_size=9, stride=2, padding=4)
        self.enc_conv2 = nn.Conv1d(16, 32, kernel_size=9, stride=2, padding=4)
        self.enc_conv3 = nn.Conv1d(32, 64, kernel_size=9, stride=2, padding=4)

        length_after = input_length
        for _ in range(3):
            length_after = int(np.ceil(length_after / 2))

        self.flatten_dim = 64 * length_after

        # bottleneck (NO cambio compresión segun dice las instrucciones)
        self.fc_enc = nn.Linear(self.flatten_dim, bottleneck_size)
        self.fc_dec = nn.Linear(bottleneck_size, self.flatten_dim)

        # decoder
        self.dec_conv1 = nn.ConvTranspose1d(64, 32, kernel_size=9, stride=2, padding=4, output_padding=1)
        self.dec_conv2 = nn.ConvTranspose1d(32, 16, kernel_size=9, stride=2, padding=4, output_padding=1)
        self.dec_conv3 = nn.ConvTranspose1d(16, 1,  kernel_size=9, stride=2, padding=4, output_padding=1)

    def encode(self, x):
        x = F.relu(self.enc_conv1(x))
        x = F.relu(self.enc_conv2(x))
        x = F.relu(self.enc_conv3(x))
        x = x.flatten(start_dim=1)
        z = self.fc_enc(x)
        return z

    def decode(self, z):
        x = self.fc_dec(z)
        x = x.view(x.size(0), 64, -1)
        x = F.relu(self.dec_conv1(x))
        x = F.relu(self.dec_conv2(x))
        x = self.dec_conv3(x)
        x = x[..., :self.input_length]
        return x

    def forward(self, x):
        z = self.encode(x)
        x_rec = self.decode(z)
        return x_rec, z

#  check
tmp = AudioAutoencoder(input_length=TARGET_LENGTH, bottleneck_size=512).to(device)
n_params = sum(p.numel() for p in tmp.parameters() if p.requires_grad)
print("AudioAutoencoder listo. Params:", f"{n_params:,}")
del tmp
```

```
AudioAutoencoder listo. Params: 131,247,041
```

In [6]:

```python
from tqdm.auto import tqdm

def train_autoencoder(
    model,
    train_loader,
    val_loader,
    n_epochs=N_EPOCHS,
    lr=1e-3,
    use_perceptual=False,
    alpha=0.1,
    project_name="lab6_autoencoder"
):
    model = model.to(device)
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)

    train_losses = []
    val_losses = []
    val_mse_list = []
    val_stft_list = []

    cpu_pct_per_epoch = []
    ram_gb_per_epoch = []
    gpu_alloc_gb_per_epoch = []
    gpu_peak_gb_per_epoch = []

    if device == "cuda":
        try:
            torch.cuda.reset_peak_memory_stats()
        except Exception:
            pass

    try:
        tracker = EmissionsTracker(project_name=project_name, log_level="error", save_to_file=True)
        tracker.start()
    except Exception:
        tracker = None

    t_global = time.time()

    pbar_epoch = tqdm(range(n_epochs), desc="Epochs", unit="epoch")
    for epoch in pbar_epoch:
        model.train()
        batch_train_losses = []

        pbar_batch = tqdm(train_loader, desc=f"Train {epoch+1}/{n_epochs}", unit="batch", leave=False)
        for batch in pbar_batch:
            x = batch["audio"].to(device, non_blocking=True).unsqueeze(1)

            optimizer.zero_grad(set_to_none=True)
            x_rec, _ = model(x)

            if use_perceptual:
                loss, _, _ = combined_loss(x_rec, x, alpha=alpha)
            else:
                loss = mse_loss(x_rec, x)

            loss.backward()
            optimizer.step()

            batch_train_losses.append(loss.item())
            pbar_batch.set_postfix({"loss": f"{loss.item():.5f}"})

        epoch_train = float(np.mean(batch_train_losses))
        train_losses.append(epoch_train)

        model.eval()
        batch_val_losses = []
        batch_val_mse = []
        batch_val_stft = []

        with torch.no_grad():
            pbar_val = tqdm(val_loader, desc=f"Val {epoch+1}/{n_epochs}", unit="batch", leave=False)
            for batch in pbar_val:
                x = batch["audio"].to(device, non_blocking=True).unsqueeze(1)
                x_rec, _ = model(x)

                v_mse = mse_loss(x_rec, x)
                batch_val_mse.append(v_mse.item())

                if use_perceptual:
                    v_loss, _, v_stft = combined_loss(x_rec, x, alpha=alpha)
                    batch_val_losses.append(v_loss.item())
                    batch_val_stft.append(v_stft.item())
                    pbar_val.set_postfix({"stft": f"{v_stft.item():.5f}", "mse": f"{v_mse.item():.5f}"})
                else:
                    batch_val_losses.append(v_mse.item())
                    pbar_val.set_postfix({"mse": f"{v_mse.item():.5f}"})

        epoch_val = float(np.mean(batch_val_losses))
        epoch_val_mse = float(np.mean(batch_val_mse))
        val_losses.append(epoch_val)
        val_mse_list.append(epoch_val_mse)

        if len(batch_val_stft) > 0:
            val_stft_list.append(float(np.mean(batch_val_stft)))

        cpu_pct, ram_gb = get_cpu_ram()
        gpu_alloc_gb, gpu_peak_gb = get_gpu_mem()

        cpu_pct_per_epoch.append(cpu_pct)
        ram_gb_per_epoch.append(ram_gb)
        gpu_alloc_gb_per_epoch.append(gpu_alloc_gb)
        gpu_peak_gb_per_epoch.append(gpu_peak_gb)

        postfix = {
            "train": f"{epoch_train:.5f}",
            "val_mse": f"{epoch_val_mse:.5f}",
            "sec": f"{time.time()-t_global:.1f}"
        }
        if cpu_pct is not None:
            postfix["cpu%"] = f"{cpu_pct:.1f}"
        if ram_gb is not None:
            postfix["ramGB"] = f"{ram_gb:.2f}"
        if gpu_peak_gb is not None:
            postfix["gpuGB"] = f"{gpu_peak_gb:.2f}"

        pbar_epoch.set_postfix(postfix)

        print(f"[{epoch+1}/{n_epochs}] train={epoch_train:.6f} val_mse={epoch_val_mse:.6f}")

    emissions_kg = None
    if tracker is not None:
        try:
            emissions_kg = tracker.stop()
        except Exception:
            emissions_kg = None

    total_time_sec = time.time() - t_global

    def _safe_max(xs):
        xs2 = [x for x in xs if x is not None]
        return float(max(xs2)) if len(xs2) > 0 else None

    def _safe_mean(xs):
        xs2 = [x for x in xs if x is not None]
        return float(np.mean(xs2)) if len(xs2) > 0 else None

    return model, {
        "train_losses": train_losses,
        "val_losses": val_losses,
        "val_mse": val_mse_list,
        "val_stft": val_stft_list if len(val_stft_list) > 0 else None,
        "emissions_kg": emissions_kg,
        "time_sec": total_time_sec,
        "time_min": total_time_sec / 60.0,
        "cpu_pct_per_epoch": cpu_pct_per_epoch,
        "ram_gb_per_epoch": ram_gb_per_epoch,
        "gpu_alloc_gb_per_epoch": gpu_alloc_gb_per_epoch,
        "gpu_peak_gb_per_epoch": gpu_peak_gb_per_epoch,
        "cpu_pct_mean": _safe_mean(cpu_pct_per_epoch),
        "ram_gb_max": _safe_max(ram_gb_per_epoch),
        "gpu_peak_gb_max": _safe_max(gpu_peak_gb_per_epoch)
    }
```

MSE

In [7]:

```python
model_mse = AudioAutoencoder(input_length=TARGET_LENGTH, bottleneck_size=512).to(device)

model_mse, results_mse = train_autoencoder(
    model=model_mse,
    train_loader=train_loader,
    val_loader=val_loader,
    n_epochs=N_EPOCHS,
    lr=1e-3,
    use_perceptual=False,
    project_name="lab6_autoencoder_mse"
)

results_mse
```

```
[codecarbon WARNING @ 21:30:48] Multiple instances of codecarbon are allowed to run at the same time.
Epochs:   0%|          | 0/50 [00:00<?, ?epoch/s]
Train 1/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 1/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[1/50] train=0.027414 val_mse=0.026258
Train 2/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 2/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[2/50] train=0.025809 val_mse=0.026239
Train 3/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 3/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[3/50] train=0.025407 val_mse=0.026241
Train 4/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 4/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[4/50] train=0.025727 val_mse=0.026243
Train 5/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 5/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[5/50] train=0.025502 val_mse=0.026237
Train 6/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 6/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[6/50] train=0.025787 val_mse=0.026204
Train 7/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 7/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[7/50] train=0.025591 val_mse=0.026236
Train 8/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 8/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[8/50] train=0.025519 val_mse=0.025114
Train 9/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 9/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[9/50] train=0.020487 val_mse=0.018299
Train 10/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 10/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[10/50] train=0.016055 val_mse=0.015871
Train 11/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 11/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[11/50] train=0.014404 val_mse=0.014460
Train 12/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 12/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[12/50] train=0.013257 val_mse=0.013390
Train 13/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 13/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[13/50] train=0.012264 val_mse=0.012692
Train 14/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 14/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[14/50] train=0.011598 val_mse=0.011878
Train 15/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 15/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[15/50] train=0.010928 val_mse=0.011279
Train 16/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 16/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[16/50] train=0.010454 val_mse=0.010851
Train 17/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 17/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[17/50] train=0.010167 val_mse=0.010571
Train 18/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 18/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[18/50] train=0.009911 val_mse=0.010318
Train 19/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 19/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[19/50] train=0.009611 val_mse=0.010170
Train 20/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 20/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[20/50] train=0.009620 val_mse=0.010089
Train 21/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 21/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[21/50] train=0.009436 val_mse=0.010028
Train 22/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 22/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[22/50] train=0.009578 val_mse=0.010021
Train 23/50:   0%|          | 0/125 [00:00<?, ?batch/s]
```

```
Val 23/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[23/50] train=0.009420 val_mse=0.009992
Train 24/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 24/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[24/50] train=0.009352 val_mse=0.009963
Train 25/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 25/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[25/50] train=0.009427 val_mse=0.010065
Train 26/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 26/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[26/50] train=0.009428 val_mse=0.009934
Train 27/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 27/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[27/50] train=0.009336 val_mse=0.009919
Train 28/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 28/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[28/50] train=0.009516 val_mse=0.009904
Train 29/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 29/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[29/50] train=0.009348 val_mse=0.009891
Train 30/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 30/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[30/50] train=0.009307 val_mse=0.009891
Train 31/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 31/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[31/50] train=0.009375 val_mse=0.009863
Train 32/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 32/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[32/50] train=0.009446 val_mse=0.009867
Train 33/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 33/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[33/50] train=0.009377 val_mse=0.009855
Train 34/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 34/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[34/50] train=0.009256 val_mse=0.009903
Train 35/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 35/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[35/50] train=0.009242 val_mse=0.009832
Train 36/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 36/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[36/50] train=0.009331 val_mse=0.009810
Train 37/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 37/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[37/50] train=0.009234 val_mse=0.009820
Train 38/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 38/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[38/50] train=0.009395 val_mse=0.009839
Train 39/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 39/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[39/50] train=0.009348 val_mse=0.009827
Train 40/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 40/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[40/50] train=0.009167 val_mse=0.009823
Train 41/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 41/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[41/50] train=0.009385 val_mse=0.009796
Train 42/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 42/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[42/50] train=0.009326 val_mse=0.009775
Train 43/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 43/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[43/50] train=0.009085 val_mse=0.009761
Train 44/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 44/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[44/50] train=0.009331 val_mse=0.009752
Train 45/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 45/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[45/50] train=0.009212 val_mse=0.009779
Train 46/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 46/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[46/50] train=0.009256 val_mse=0.009749
Train 47/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 47/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[47/50] train=0.009272 val_mse=0.009826
Train 48/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 48/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[48/50] train=0.009190 val_mse=0.009734
Train 49/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 49/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[49/50] train=0.009232 val_mse=0.009752
Train 50/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 50/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[50/50] train=0.009312 val_mse=0.009721
```

```
Out[7]: {'train_losses': [0.027414493784308432,
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```
In [8]: #N_EPOCHS = 8
```

PERCEPTUAL LOSS

```
In [9]: # entrenamiento con perdida perceptual
model_perc = AudioAutoencoder(
    input_length=TARGET_LENGTH,
    bottleneck_size=512
).to(device)

model_perc, results_perc = train_autoencoder(
    model=model_perc,
    train_loader=train_loader,
    val_loader=val_loader,
    n_epochs=N_EPOCHS_PERC,
    lr=1e-3,
    use_perceptual=True,
    alpha=0.01,
    project_name="lab6_autoencoder_perc"
)

results_perc
```

```
Epochs:   0%|          | 0/50 [00:00<?, ?epoch/s]
Train 1/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 1/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[1/50] train=0.050706 val_mse=0.026619
Train 2/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 2/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[2/50] train=0.044708 val_mse=0.026494
Train 3/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 3/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[3/50] train=0.043845 val_mse=0.027060
Train 4/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 4/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[4/50] train=0.043638 val_mse=0.026823
Train 5/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 5/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[5/50] train=0.043060 val_mse=0.027000
Train 6/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 6/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[6/50] train=0.042756 val_mse=0.026626
Train 7/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 7/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[7/50] train=0.042555 val_mse=0.026753
Train 8/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 8/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[8/50] train=0.042244 val_mse=0.026661
Train 9/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 9/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[9/50] train=0.041956 val_mse=0.026593
Train 10/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 10/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[10/50] train=0.042068 val_mse=0.026682
Train 11/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 11/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[11/50] train=0.041753 val_mse=0.026700
Train 12/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 12/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[12/50] train=0.042299 val_mse=0.026821
Train 13/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 13/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[13/50] train=0.041581 val_mse=0.026505
Train 14/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 14/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[14/50] train=0.041806 val_mse=0.026674
Train 15/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 15/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[15/50] train=0.042043 val_mse=0.026864
Train 16/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 16/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[16/50] train=0.041875 val_mse=0.027651
Train 17/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 17/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[17/50] train=0.042101 val_mse=0.026973
Train 18/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 18/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[18/50] train=0.041969 val_mse=0.026744
Train 19/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 19/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[19/50] train=0.041201 val_mse=0.026810
Train 20/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 20/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[20/50] train=0.041033 val_mse=0.026523
Train 21/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 21/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[21/50] train=0.041513 val_mse=0.028340
Train 22/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 22/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[22/50] train=0.041429 val_mse=0.026664
Train 23/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 23/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[23/50] train=0.042223 val_mse=0.026825
Train 24/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 24/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[24/50] train=0.041166 val_mse=0.026594
Train 25/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 25/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[25/50] train=0.041003 val_mse=0.026712
Train 26/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 26/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[26/50] train=0.041549 val_mse=0.026624
Train 27/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 27/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[27/50] train=0.040939 val_mse=0.026434
Train 28/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 28/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[28/50] train=0.040763 val_mse=0.025821
Train 29/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 29/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[29/50] train=0.039205 val_mse=0.023346
Train 30/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 30/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[30/50] train=0.037577 val_mse=0.022550
[30/50] train=0.037577 val_mse=0.022550
Train 31/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 31/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[31/50] train=0.037880 val_mse=0.021507
Train 32/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 32/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[32/50] train=0.036194 val_mse=0.020276
Train 33/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 33/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[33/50] train=0.035117 val_mse=0.019518
Train 34/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 34/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[34/50] train=0.034388 val_mse=0.019013
Train 35/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 35/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[35/50] train=0.034516 val_mse=0.018965
Train 36/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 36/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[36/50] train=0.033473 val_mse=0.018119
Train 37/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 37/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[37/50] train=0.033241 val_mse=0.017969
Train 38/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 38/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[38/50] train=0.032187 val_mse=0.017483
Train 39/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 39/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[39/50] train=0.031923 val_mse=0.017027
Train 40/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 40/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[40/50] train=0.031817 val_mse=0.017625
Train 41/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 41/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[41/50] train=0.031650 val_mse=0.016750
Train 42/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 42/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[42/50] train=0.031288 val_mse=0.016787
Train 43/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 43/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[43/50] train=0.030800 val_mse=0.016208
Train 44/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 44/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[44/50] train=0.031079 val_mse=0.016432
Train 45/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 45/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[45/50] train=0.031016 val_mse=0.015349
Train 46/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 46/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[46/50] train=0.031936 val_mse=0.016239
Train 47/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 47/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[47/50] train=0.031529 val_mse=0.016951
Train 48/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 48/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[48/50] train=0.030401 val_mse=0.015707
Train 49/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 49/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[49/50] train=0.029746 val_mse=0.015680
Train 50/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 50/50:   0%|          | 0/16 [00:00<?, ?batch/s]
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```

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```
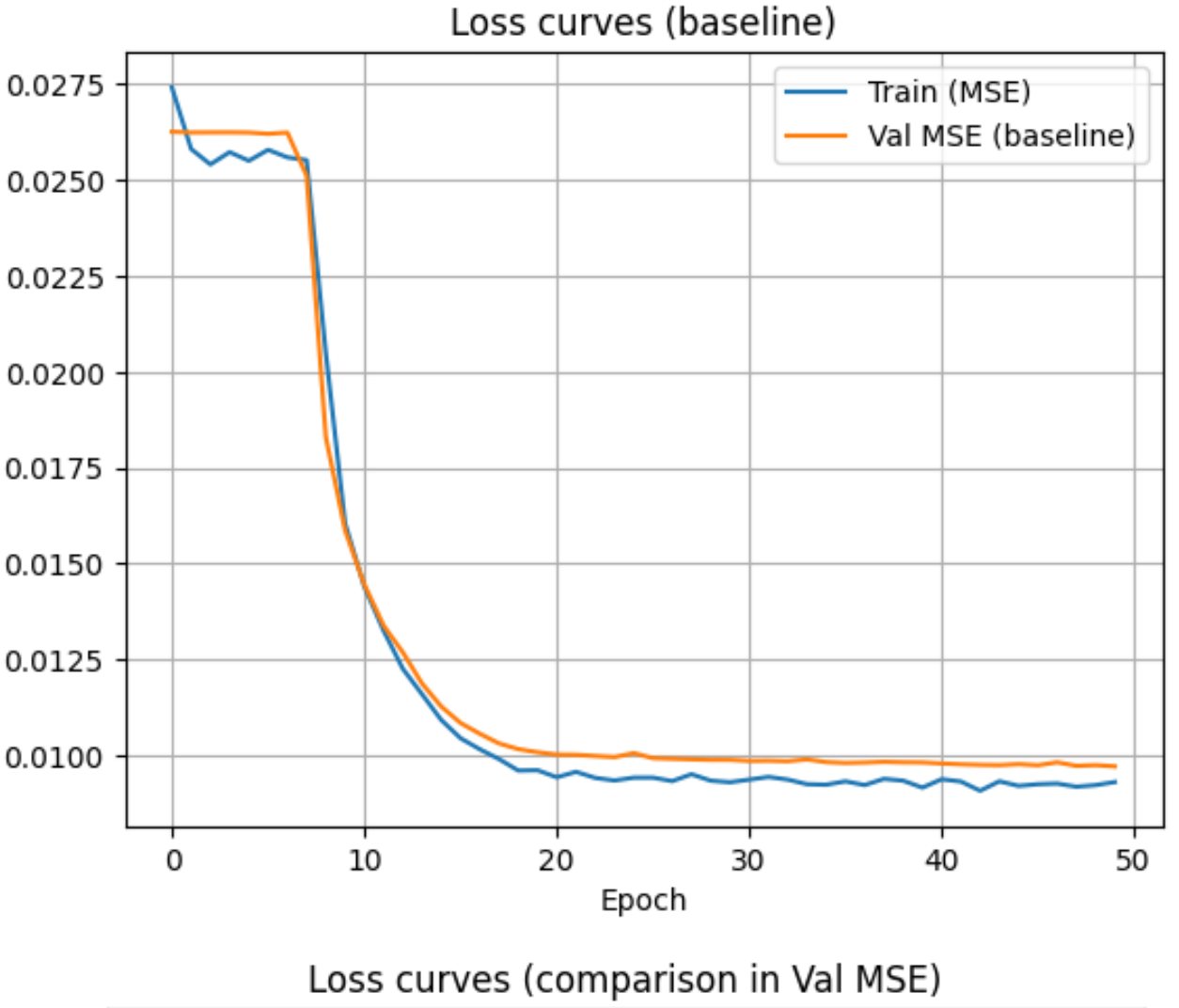
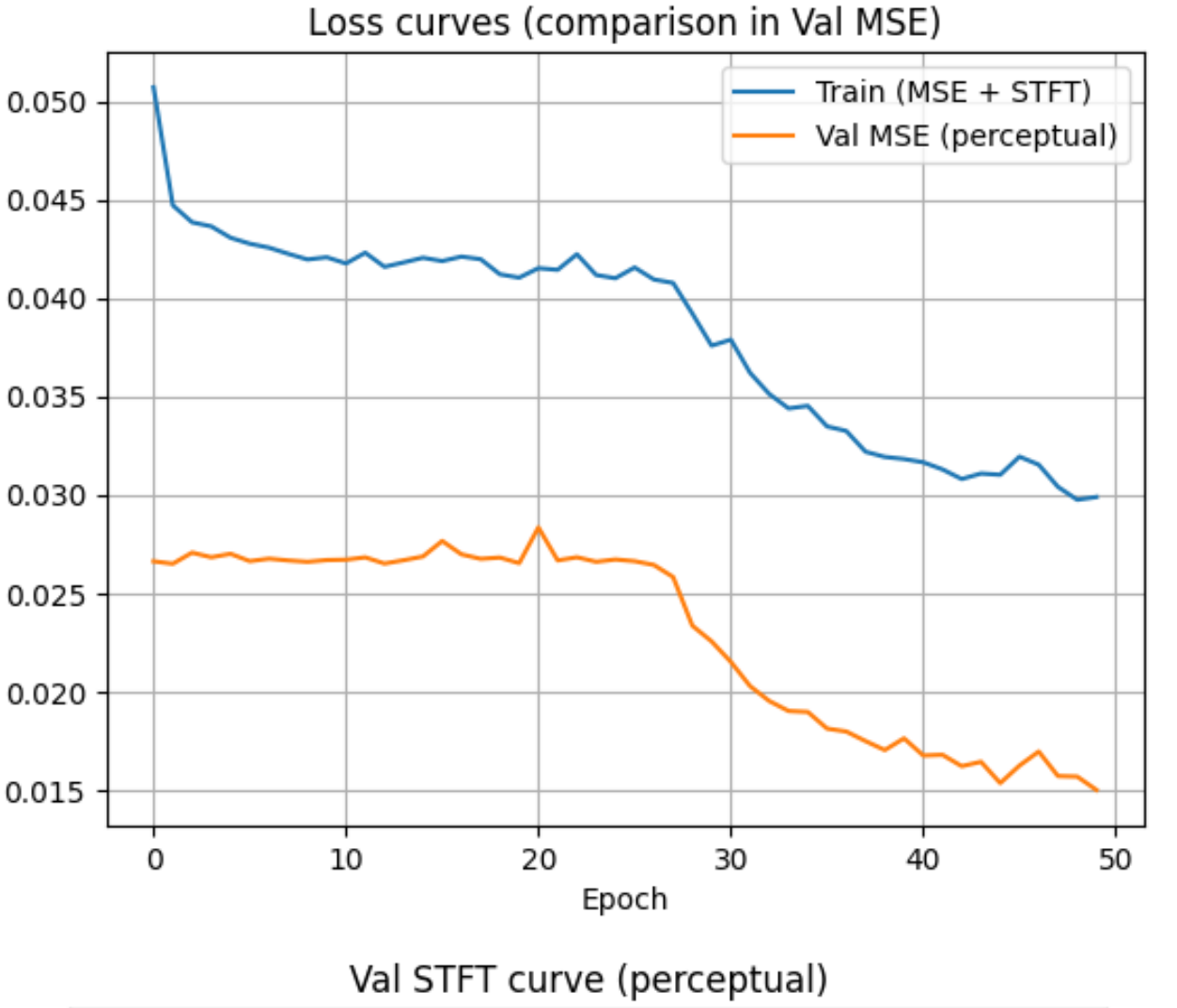
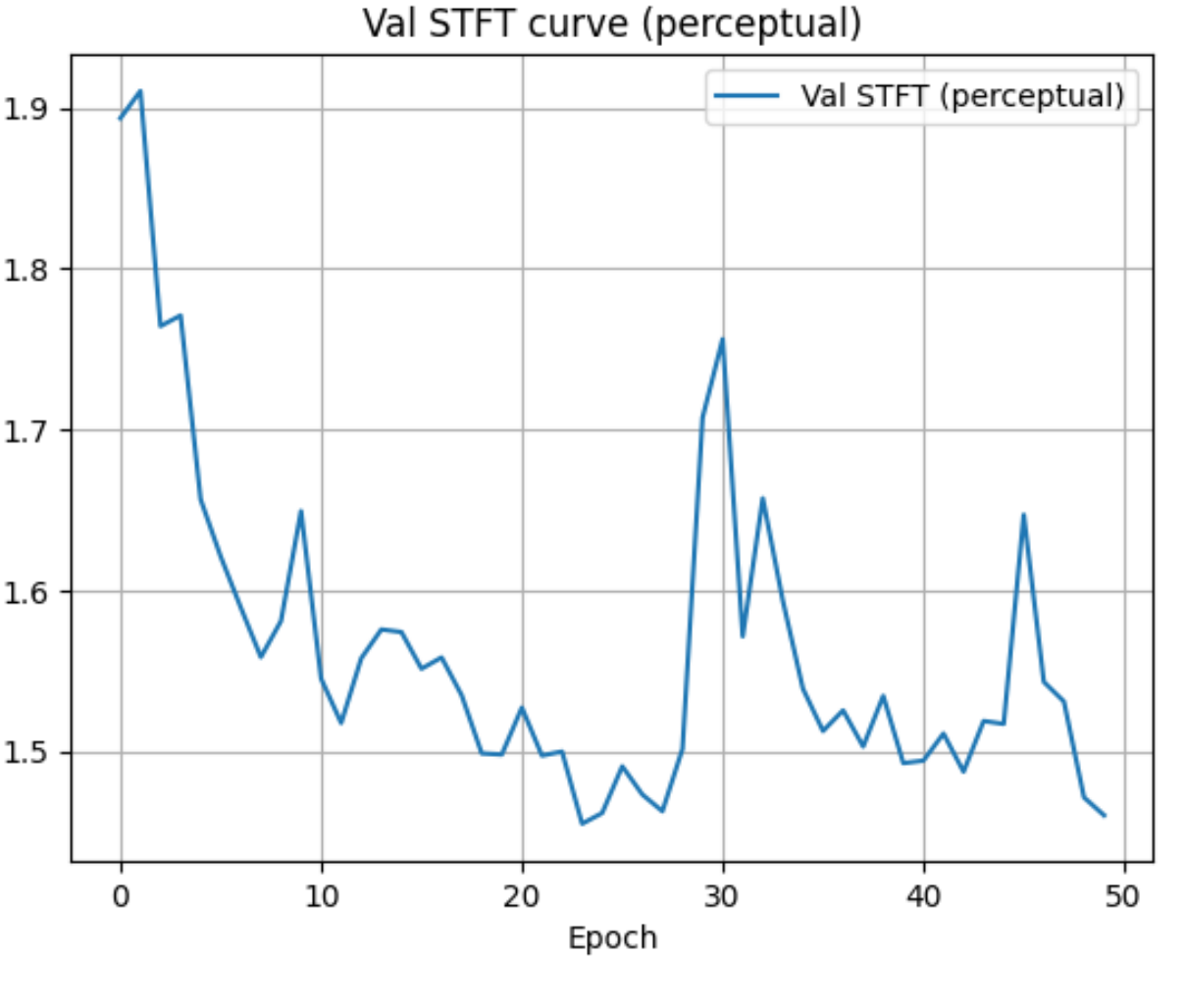

In [10]:

```python
# grafica de perdidas

plt.figure()
plt.plot(results_mse["train_losses"], label="Train (MSE)")
plt.plot(results_mse["val_mse"], label="Val MSE (baseline)")
plt.xlabel("Epoch")
plt.ylabel("Loss / MSE")
plt.legend()
plt.title("Loss curves (baseline)")
plt.grid(True)
plt.show()

if "results_perc" in globals():
    plt.figure()
    plt.plot(results_perc["train_losses"], label="Train (MSE + STFT)")
    plt.plot(results_perc["val_mse"], label="Val MSE (perceptual)")
    plt.xlabel("Epoch")
    plt.ylabel("Loss / MSE")
    plt.legend()
    plt.title("Loss curves (comparison in Val MSE)")
    plt.grid(True)
    plt.show()

    # grafica extra: solo para ver STFT val del perceptual

    if results_perc["val_stft"] is not None:
        plt.figure()
        plt.plot(results_perc["val_stft"], label="Val STFT (perceptual)")
        plt.xlabel("Epoch")
        plt.ylabel("STFT loss")
        plt.legend()
        plt.title("Val STFT curve (perceptual)")
        plt.grid(True)
        plt.show()
```







QUANTITATIVE PERFRMANCE

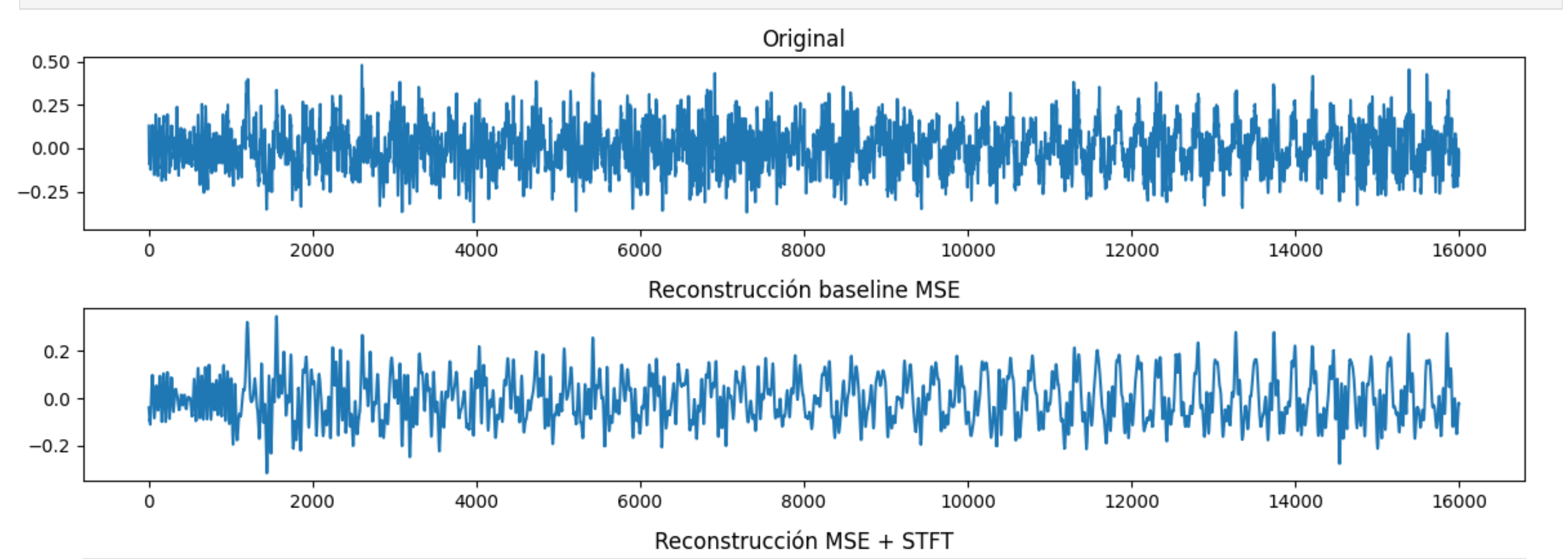
The algorithm/model based on MSE achieves a lower reconstruction error, in terms of sample wise loss (as expected, due to its optimization objective). The perceptual loss model doesn't outperform the baseline in MSE metrics but it achieves lower perceptual loss values (multi resolution STFT). This indicates an improved spectral similarity between original and reconstructed signals.

These results highight the trade off between waveform accuracy and percetual quality in audio reconstrution tasks.

WAVEFORM PLOTTING

In [ ]:

```python
from IPython.display import Audio, display

batch = next(iter(test_loader))
x = batch["audio"][:4].float().to(device).unsqueeze(1)

model_mse.eval()
model_perc.eval()

with torch.no_grad():
    x_rec_mse, _ = model_mse(x)
    x_rec_perc, _ = model_perc(x)

# waveform original vs reconstruida
idx = 0
orig = x[idx].detach().cpu().squeeze().numpy()
rec_mse = x_rec_mse[idx].detach().cpu().squeeze().numpy()
rec_perc = x_rec_perc[idx].detach().cpu().squeeze().numpy()

plt.figure(figsize=(12, 6))
plt.subplot(3, 1, 1)
plt.plot(orig)
plt.title("Original")

plt.subplot(3, 1, 2)
plt.plot(rec_mse)
plt.title("Reconstrucción baseline MSE")

plt.subplot(3, 1, 3)
plt.plot(rec_perc)
plt.title("Reconstrucción MSE + STFT")
plt.tight_layout()
plt.show()
```



12 EXAMPLES FOR ANALYSIS

In [12]:

```python
from IPython.display import Audio, display
import numpy as np
import torch

def rms(x, eps=1e-8):
    return np.sqrt(np.mean(x**2) + eps)

def match_rms(y, ref, eps=1e-8):
    ry = rms(y, eps)
    rr = rms(ref, eps)
    return y * (rr / (ry + eps))

def to_audio(x_t):
    return x_t.detach().cpu().squeeze().numpy()

def show_examples(model_a, model_b, loader, n=12, seed=0):
    g = torch.Generator()
    g.manual_seed(seed)

    model_a.eval()
    model_b.eval()

    batch = next(iter(loader))
    x = batch["audio"].float()

    if x.shape[0] < n:
        n = x.shape[0]

    idxs = torch.randperm(x.shape[0], generator=g)[:n].tolist()
    x_sel = x[idxs].to(device).unsqueeze(1)

    with torch.no_grad():
        ra, _ = model_a(x_sel)
        rb, _ = model_b(x_sel)

    for i in range(n):
        orig = to_audio(x_sel[i])
        a = to_audio(ra[i])
        b = to_audio(rb[i])

        a = match_rms(a, orig)
        b = match_rms(b, orig)

        a = np.clip(a, -1.0, 1.0)
        b = np.clip(b, -1.0, 1.0)

        print(f"Example {i+1}/{n}")
        print("Original:")
        display(Audio(orig, rate=SAMPLE_RATE))
        print("Recon (MSE):")
        display(Audio(a, rate=SAMPLE_RATE))
        print("Recon (Perceptual):")
        display(Audio(b, rate=SAMPLE_RATE))
        print("-" * 60)

show_examples(model_mse, model_perc, test_loader, n=12, seed=42)
```

```
Example 1/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 2/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 3/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 4/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 5/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 6/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 7/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 8/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 9/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 10/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 11/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 12/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
```

ABOUT THE TECHNIQUES USED IN THIS LAB

LONG AUDIOS STRATEGY

Handled variable and length audio by enforcing a fixed input length (1 second):

- Training: random crop (seeking to add temporal augmentation and improve generalization).
- Validation/Test: centered crop (to ensure consistent evaluation).

This strategy is expected to allow training the model on MTAT clips with different durations while keeping batch shapes fixed.

RECONSTRUCTION IMPROVEMENT (PERCEPTUAL LOSS)

- Trained a second model using a perceptual loss based on multi resolution STFT trying to combine it with MSE.
- The goal is to improve perceived spectral quality of reconstructions (timbre, detail), even if there's not a notable improvement in MSE.

AUDIO RECONSTRUCTION RESULTS

The baseline autoencoder trained with MSE produces reconstructions that can be considered indetifiable and preserves overall a temporal structure (and rhythm) of the original audio.

However, the reconstructed signals sound perceptually muffled, with reduced brightness and less defined transients which translates to a boxed or damped audio character.

In contrast, the perceptual-loss model emphasizes low frequency energy and rhytmic components. The reconstructed audio preserves bass content and structures with identifiable beat in a more effective way; all of this despite high frequency details and spectral brightness remains attenuated. This behavior, reflects the trend for spectral loss to prioritize global spectral similarity over temporal details.

RESULTS, DICUSSION, RESOURCES AND CONCLUSION

- In the notebook you can find: reported training and validation curves and compared both models using a common metric (Validation MSE).
- For the perceptual model, also tracked the validation perceptual/STFT loss, mainly as an auxiliary metric.
- Measured total training time and estimated CO2 emissions using CodeCarbon for resource (and) environmental reporting.

In [13]:

```python
# resumen final de resultados y emisiones
def print_summary(name, res):
    print(f"=== {name} ===")

    # métrica comparable entre modelos
    if "val_mse" in res and res["val_mse"] is not None:
        print(f"Best Val MSE:  {min(res['val_mse']):.6f}")

    # métrica perceptual (solo aplica si existe)
    if "val_stft" in res and res["val_stft"] is not None:
        print(f"Best Val STFT: {min(res['val_stft']):.6f}")

    print(f"Total time:  {res['time_min']:.2f} min")
    print(f"Total CO2 Emissions: {res['emissions_kg']:.6f} kg")
    print()

print_summary("Baseline MSE", results_mse)
print_summary("MSE + STFT",   results_perc)


print("CPU mean (%):", results_mse.get("cpu_pct_mean"), "| RAM max (GB):", results_mse.get("ram_gb_max"), "| GPU peak (GB):", results_mse.get("gpu_peak_
print("CPU mean (%):", results_perc.get("cpu_pct_mean"), "| RAM max (GB):", results_perc.get("ram_gb_max"), "| GPU peak (GB):", results_perc.get("gpu_pea
```

```
=== Baseline MSE ===
Best Val MSE:  0.009721
Total time:  35.15 min
Total CO2 Emissions: 0.045078 kg

=== MSE + STFT ===
Best Val MSE:  0.015000
Best Val STFT: 1.455055
Total time:  40.66 min
Total CO2 Emissions: 0.052221 kg

CPU mean (%): 0.0 | RAM max (GB): 6.573619842529297 | GPU peak (GB): 2.4769086837768555
CPU mean (%): 0.0 | RAM max (GB): 6.8452301025390625 | GPU peak (GB): 3.449263572692871
```

DISCUSSION

Both models are successful in learning a compressed latent representation of audio signals. However, the convolutional architecture for the autoencoder, combined with a fixed bottleneck size, limits the ability to reconstruct content with high frequencies in an accurate way. Same happens with sharp transients.

The MSE model favors smooth reconstructions that minimize point wise error but at the expense of perceptual clarity, hence why it sounds like it was in a box or like when you hear music in a room nextdoors. The perceptual loss model focuses the optimization towards spectral characteristics. This improves the representation of low frequencies and also rhythm but at the expense of waveform precision plus increasing computational cost.

These observations (in multiple executions of the code) were consisten with established behavior in audio autoencoding and confirms the trade offs between perceptual quality and objective loss functions plus model complexity.

COMPUTATIONAL RESOURCES AND ENVIRONMENTAL IMPACT

In addition to reconstruction performance: computational resources and environmental impact were measured during the whole process.

- The baseline MSE model required less training time and lower GPU memory usage. Consequently, this results in lower CO2 emissions.

- The perceptual loss model required longer training time and higher computational resources because of the multi resolution spectral loss computation. This lead to an increase in energy consumption and also CO2 emissions. This highlights the importance of considering environmental costs when designing and evaluating machine learning models. Also, in a commercial point of view it comes with a higher monetary cost which might not help in the profitability of a system.

CONCLUSION

This laboratory shows the different loss functions and how they lead to distinct trade offs in audio autoencoding.

While MSE model provides stable and intelligible reconstructions, with lower computational cost, the perceptual loss model prioritizes spectral similarity and rhythmic structure at the expense of waveform accuracy and efficiency.

The results shown in the notebook confirm that perceptual improvements in audio reconstruction usually come with increased computational and environmental costs. This emphasizes the need for balanced model designs, depending on the needs.